

ETAS-Inspired Spatio-Temporal Convolutional (STC) Model for Next-Day Earthquake Forecasting

Chengxiang Zhan¹, Shichen Gao¹, Ying Zhang¹, Jiawei Li², and Qingyan Meng¹

Abstract—Research on integrating statistical knowledge into deep learning models for earthquake forecasting has been limited. Traditional deep learning models require extensive parameter learning from scratch. This study proposes a spatio-temporal convolutional (STC) model that incorporates spatio-temporal decay prior knowledge derived from the epidemic-type aftershock sequence (ETAS) into a convolutional kernel. This allows the STC model to have the prototype to learn the pattern of mainshocks to trigger aftershocks at the beginning of training, with only four neurons to fine-tune it. In California, the STC and the ETAS model are conducted for forecasting next-day earthquakes with magnitudes of $M \geq 3$, $M \geq 4$, and $M \geq 5$. Both performances were assessed using the receiver operating characteristic (ROC) curve, the precision–recall (PR) curve, and the parimutuel gambling score (PGS). The evaluation results indicate that the STC model surpasses ETAS in forecasting next-day earthquakes not accidental. Furthermore, our analysis suggests that including earthquakes below the complete magnitudes can enhance the STC model’s classification performance, as small earthquakes also contain information about future earthquakes.

Index Terms—Deep learning, earthquake forecasting, statistical seismology.

I. INTRODUCTION

SHORT-TERM probabilistic earthquake forecasting remains a focal point of intense research, bridging the realms of earthquake hazard characterization and risk mitigation [1]. This endeavor poses a formidable challenge to geophysics, statistical seismology, and the burgeoning field of artificial intelligence (AI), captivated the scientific community for decades.

In statistical seismology, the prevailing approach is to model earthquake sequences as spatio-temporal point processes [2].

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Chengxiang Zhan and Shichen Gao are with the School of Science, China University of Geosciences (Beijing), Beijing 100083, China (e-mail: zhanxc@email.cugb.edu.cn; gsc2039@cugb.edu.cn).

Ying Zhang is with the School of Automation and Electrical Engineering, University of Science and Technology Beijing, Beijing 100083, China (e-mail: zhangying2016@radi.ac.cn).

Jiawei Li is with the Institute of Risk Analysis, Prediction and Management (Risks-X), Academy for Advanced Interdisciplinary Studies, Southern University of Science and Technology (SUSTech), Shenzhen 518055, China (e-mail: lijw3@sustech.edu.cn).

Qingyan Meng is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100049, China (e-mail: mengqy@radi.ac.cn).

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Self-exciting processes, known as Hawkes processes [3], [4], are particularly favored for their capacity to encapsulate the temporal, spatial, and network dynamics of earthquakes [2], [5]. Established empirical laws, such as Gutenberg–Richter law [6] and Omori’s law [7], elucidate the dependencies between earthquakes. Building upon this statistical foundation, the epidemic-type aftershock sequence (ETAS) [8], [9], [10], [11] has emerged as a pivotal tool for forecasting earthquakes. This model, which has been refined over a few decades, captures the triggering mechanisms of aftershocks through a spatio-temporal exponential decay diffusion kernel. Despite its widespread application and success, the ETAS model has limitations. First, small earthquakes below the complete magnitude also contain valuable information for earthquake prediction, but ETAS incorporating these earthquakes into the simulation would instead lead to biased parameter estimates [12], [13]. Second, the kernel function is spatially isotropic and cannot capture the complex spatial relationships between main and aftershocks [14]. Moreover, calibrating the model’s parameters necessitates the analysis of extensive historical data, a time-consuming endeavor.

AI, with its prowess in discerning patterns within vast datasets, has become an indispensable ally in earthquake prediction. Over recent years, AI-driven approaches, such as neural networks (NNs) [15], [16], deep NNs (DNNs) [17], convolutional NNs (CNNs) [18], recurrent NNs (RNNs) [19], long-short term memory (LSTM) network [20], a combination of CNN and LSTM [21], [22], [23], and attention mechanism [24], [25], have been employed with promising results. The advantage of these models is that they can fit complex nonlinear relationships based on many trainable parameters and autonomous learning capabilities, potentially surpassing ETAS in predictive performance [26]. However, based on machine learning evaluation metrics, these AI models can often be considered better than other simple machine learning approaches, but fewer research on whether they outperform well-established physical or statistical models [27]. In addition, the training on these models usually relies on a large amount of input data, which is often limited in observational information on earthquakes, especially large earthquakes. Furthermore, the lack of interpretability inherent in deep learning models further complicates the understanding of their forecasting and the insights they provide [28].

The strengths and weaknesses of both statistical seismology and AI are well-documented. Integrating physical knowledge into the training data, hypothesis space, and procedures of deep learning models has been a recent focus [28]. Notable

efforts, such as the convolutional LSTM (CL)-ETAS model by Zhang et al. [29], have leveraged Conv-LSTM to assimilate knowledge from historical data and the ETAS model. Liu et al. [30] incorporated spatio-temporal prior knowledge into DNNs using limited historical earthquake samples to predict earthquake magnitude and epicenter. Nicolis et al. [31] used earthquake intensity output from ETAS as an input to LSTM and CNN, predicting the intensity and probability of earthquakes for the next day, respectively. In this way, earthquake forecasting models can take advantage of specific statistical features, namely, temporal and spatial correlation, not easily obtained from a complete NN model [5]. Furthermore, Zlydenko et al. [32] proposed an encoder–decoder neural-based model for predicting earthquakes. The model can capture complex and anisotropic spatial relationships, and its performance can be further improved by including small earthquakes. Zhu et al. [33] proposed the neural embedding spatio-temporal model to describe the spatial heterogeneity of conditional intensity while maintaining interpretability. These studies have partially addressed the spatial isotropy issue in ETAS and the interpretability challenge in deep learning models. However, to the best of our knowledge, the application of statistical prior knowledge to deep learning models is under-explored, with most models trained from scratch, leading to significant uncertainty and potential overfitting or underfitting.

We proposed a novel spatio-temporal convolutional (STC) model for earthquake prediction in California. The main innovation of the STC model is the spatio-temporal convolutional kernel, derived from four neurons, which integrates spatio-temporal decay knowledge without the need for extensive trainable variables. This approach mitigates the risk of overfitting and underfitting, particularly in the context of limited sample sizes. While resembling the fine-tuning trick often employed in deep learning, which involves a pretrained model with a certain capability obtained in some other way, it is not identical. The dataset is partitioned into three sets: training, validation, and test to investigate the robustness of the STC model. The output of the STC model is the expected number of earthquakes with magnitude of $M \geq 3$, $M \geq 4$, and $M \geq 5$ occurring in each spatio-temporal bin ($0.1^\circ \times 0.1^\circ \times 1$ day). Then, the expected number within each spatio-temporal bin is converted to the probability of at least one earthquake occurring, assuming that earthquake occurrences follow a Poisson distribution.

The STC model is constructed by incorporating the a priori knowledge of aftershock decay patterns from the ETAS model into the deep learning approach. Therefore, the methodological details and earthquake forecasting performance of the STC model in conjunction with the ETAS model will be subjected to a comparative analysis. To enhance the clarity and readability of this article, the direct comparison of the STC model to a complete deep learning model has been eschewed. Instead, in the future, it would be more fruitful to develop a deep learning model that can surpass the ETAS model and elucidate the specific insights that the model can glean. In this article, we first introduce the study area and the preprocessing of the data. Second, how the STC model is inspired by ETAS will be described in detail in Section III. Then, the classification and probability forecasting performance of the STC and ETAS are

compared using the receiver operating characteristic (ROC) curve, precision–recall (PR) curve, and parimutuel gambling score (PGS). Moreover, the gain of using small-magnitude earthquakes as input to the STC model is explored.

II. STUDY AREA AND DATASET

A. Study Area

In California for example, the earthquake forecasting performance of the STC and ETAS models was evaluated and analyzed. The foundational dataset for our research is the earthquake catalog ($M \geq 0$, $40 \text{ km} \geq \text{depth} > 0$), which was obtained from the Advanced National Seismic System (ANSS) in California between January 1, 1990 and December 31, 2020. The spatial extent of this study area is the regional earthquake likelihood models (RELM) polygon proposed by Schorlemmer and Gerstenberger [34], with the spatial distribution of earthquakes illustrated in Fig. 1(a). Ensuring the completeness of the earthquake catalog is pivotal for parameter fitting in ETAS models [35]. The magnitude threshold (M_c), derived from the frequency–magnitude distribution following the Gutenberg–Richter law [6], is often applied to address completeness issues. The global empirical frequency–magnitude distribution of the earthquake catalog is shown in Fig. 1(b), with the M_c estimated to be 2.1 using the approach proposed by Clauset et al. [36], as indicated by the black solid line. The temporal increase in recorded small-magnitude earthquakes is evident in Fig. 1(c). Nandan et al. [35] indicated that M_c is spatially variable and established that $M_c = 3$ is almost applicable to the whole of California. For parameter calibration of the ETAS model, only earthquakes ($M \geq 3$) were considered. In contrast, the STC model was trained, validated, and tested using a variable magnitude threshold ranging from 0 to 3, inclusive.

B. Dataset

The whole time–space volume in California was divided into discrete time–space bins and a 3-D discrete dataset was constructed, including input data, true number, and binary label, with the shape of [11 323 (days), 125 (0.1°), and 133 (0.1°)], each dimension representing time, longitude, and latitude, respectively. The spatial scale is $0.1^\circ \times 0.1^\circ$, and the temporal scale is one day. The time–space bin is represented as $[t, x, y]$, where the space–time range cover by this bin is from $x - (x_{\text{bin}}/2)$ to $x + (x_{\text{bin}}/2)$ on the x -axis, from $y - (y_{\text{bin}}/2)$ to $y + (y_{\text{bin}}/2)$ on the y -axis, and from $t - t_{\text{bin}}$ to t on the time axis. x_{bin} , y_{bin} , and t_{bin} are the widths of time–space bins on the x -, y -, and t -axes, respectively. By applying the rules mentioned above, we convert the earthquake catalog $\text{eq}_i = \{x_i, y_i, t_i, m_i\}$ to input data $I[t, x, y]$

$$I[t, x, y] = \sum_{j: \text{eq}_j \in [t, x, y]} e^{m_j}.$$

The true number, $N[t, x, y, m]$, denotes the count of earthquakes above magnitude threshold (m) in the spatio-temporal bin $[t, x, y]$. Moreover, for evaluating the STC and ETAS models using ROC, PR curves, and PGS, $L[t, x, y, m]$ represents the binary label (“1” or “0”), signed “1” if at least one

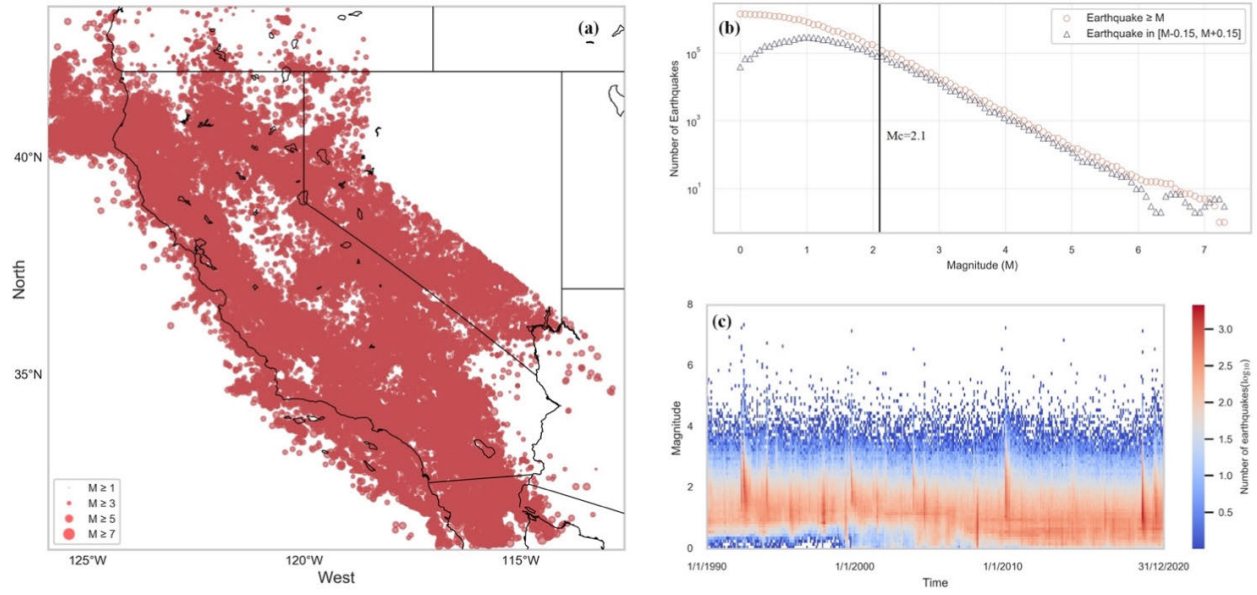


Fig. 1. (a) Spatial distribution of earthquakes ($M \geq 0$) between January 1, 1990 and December 31, 2020. (b) Frequency-magnitude distribution of earthquakes: orange circles represent earthquakes with a magnitude larger than M , gray triangles show the number of earthquakes in magnitude bins of 0.15 units, and the black solid line indicates the global completeness magnitude ($M_c = 2.1$). (c) Earthquake number over time from January 1, 1990 to December 31, 2020 (the resolution is 30 days).

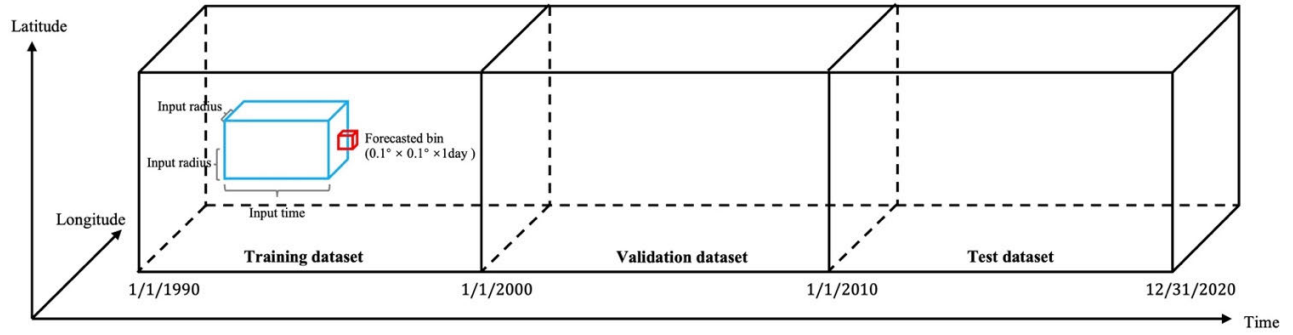


Fig. 2. Schematic for the dataset. The three black cubes, from left to right, represent the training set, validation set, and test set chronologically. The example of the input data and prediction window is represented by the blue and red cubes, respectively.

earthquake above magnitude threshold (m) occurs within it; otherwise, it is signed “0.”

For example, the red outline in Fig. 2 represents a spatio-temporal bin to be forecast, while the blue outline represents a spatio-temporal range (Ω) for the calculation of the triggered component for the red outline. The flexibility of Ω is crucial; too small, valuable data may be omitted; and too large, computational efficiency is compromised without gain. To investigate the impact of spatial and temporal input ranges on the STC model’s performance, we tested a range of input times (10, 20, 30, 40, 50, 60, 70, 80, and 90 days) and radius (0.3°, 0.6°, 0.9°, 1.2°, 1.5°, 1.8°, and 2.1°).

III. METHODOLOGY

A. Brief Overview of the ETAS Model

The ETAS model, which is designed based on Hawkes process in continuous time-space, offers a comprehensive framework for understanding the space-time clustering features of earthquakes. This model is structured around the conditional intensity function, which accounts for the history of observed earthquakes. The conditional intensity, $\lambda(t, x, y, m|H_t)$, represents the expected rate of occurrences of

earthquakes with specific time, location, and magnitude given the history H_t . Mathematically, it is expressed as (3) in Table I. The joint conditional intensity function $\lambda(t, x, y, m|H_t)$ can be divided into its marginal intensity function $S(m)$ and conditional intensity function $\lambda(t, x, y)$, as shown in (5). Also, the conditional intensity function $\lambda(t, x, y)$ can be formulated as Hawkes process in (9), where $\mu(x, y)$ denotes the background intensity, reflecting the baseline seismic activity that is independent of time. The term $k(m_i)$ specifies the expected number of events triggered by an earthquake of magnitude m_i , and $g(t - t_i, x - x_i, y - y_i, m_i)$ is the spatio-temporal distribution function of triggered earthquakes. In this work, the function of the ETAS model was developed by Nandan et al. [35] and [37]. We will detail these functions in Section III-C.

B. Introduction of the STC Model

In the STC model, the whole time-space volume was divided into discrete time-space bins and attempted to provide an integral of the conditional intensity of the Hawkes process in that bin with a convolutional kernel. According to the definition in the point process, the integral of the conditional intensity in time-space bin represents the expected number

TABLE I
CORRESPONDENCE BETWEEN THE ETAS AND STC MODEL

	ETAS		STC	
Calculation way	Hawkes point process	(1)	Discrete convolutional operation	(2)
Definition	$\lambda(t, x, y, m \mathcal{H}_t) dt dx dy dM = E(\mathcal{N}(dt dx dy dM) \mathcal{H}_t)$	(3)	$\Lambda[t, x, y, m \mathcal{H}_t] = E(\mathcal{N}[t, x, y, m] \mathcal{H}_t)$	(4)
Marginal intensity	$\lambda(t, x, y, m \mathcal{H}_t) = S(m) \lambda(t, x, y \mathcal{H}_t)$	(5)	$\Lambda[t, x, y, m \mathcal{H}_t] = J(m) \Lambda[t, x, y \mathcal{H}_t]$	(6)
Input volume	$\mathcal{H}_t = \{(t_i, x_i, y_i, m_i) : t_i < t\}$	(7)	$\mathcal{H}_t = \{(t_i, x_i, y_i, m_i) : t_i, x_i, y_i, m_i \in \Omega\}$	(8)
Conditional intensity function or its integral	$\lambda(t, x, y) = \mu(x, y) + \sum_{i: t_i < t} k(m_i) g(t - t_i, x - x_i, y - y_i, m_i)$	(9)	$\Lambda[t, x, y] = \beta \mu[x, y] + \alpha ((I * C)[t, x, y])$ $= \beta \mu[x, y] + \alpha \sum_i \sum_j \sum_k I[i, j, k] \cdot C[t - i, x - j, y - k]$	(10)
Background	$\mu(x, y)$	(11)	$\mu[x, y]$	(12)
Capacity to produce aftershocks	$k(m_i) = K e^{a(m_i - M_0)}$	(13)	$I[t, x, y] = \sum_{j: q_j \in [t, x, y]} e^{m_j}$	(14)
kernel function	$g(t - t_i, x - x_i, y - y_i, m_i) = \frac{1}{(t - t_i + c)^{1+\omega} ((x - x_i)^2 + (y - y_i)^2 + d e^{\gamma(m_i - M_0)})^{1+\rho}}$	(15)	$C[t - i, x - j, y - k] = \frac{1}{(t - i + c)^p ((x - j)^2 + (y - k)^2 + d)^q}$	(16)
Optimization objectives	$\theta = \{\mu, K, a, c, \omega, d, \gamma, \rho\}$	(17)	$\theta = \{\alpha, \beta, d, q, c, p\}$	(18)
Optimization function	$I(\theta) = \sum_t \log(\lambda(t, x, y)) - \int_0^T \iint_S \lambda(t, x, y) dx dy dt$	(19)	$loss = \text{mean}(\Lambda[t, x, y, m] - \mathcal{N}[t, x, y, m])^2$	(20)
Probability conversion	Catalog simulation	(21)	$P_{STC}[t, x, y, m] = 1 - e^{-\Lambda[t, x, y, m]}$	(22)

of events occurring in that bin. Therefore, the output of the STC model $\Lambda[t, x, y, m | H_t]$ represents the expected number of earthquakes with magnitude equal or greater than m in bin $[t, x, y]$, as shown in (4).

Similarly, in (6), the integral of the joint conditional intensity function $\Lambda[t, x, y, m | H_t]$ can be divided into its marginal intensity function $J(m)$ and the integral of the conditional density function $\Lambda[t, x, y | H_t]$. Also, $\Lambda[t, x, y | H_t]$ can be formulated as a discrete convolutional operation in (10). Here, the initial term on the right-hand side of the equation denotes the expected number of background earthquakes in the discrete space bin $[x, y]$, and the second term convolution operator represents the expected number of triggered earthquakes in the time-space bin $[t, x, y]$, where $I[i, j, k]$ indicates the input data matrix that we described in Section II-B and $C[t - i, x - j, y - k]$ represents the convolutional kernel that we will describe it in Section III-C. α and β are two neurons used to weight the expectation of triggered earthquakes and background earthquakes in the STC model, respectively, with an initial value of 1.

C. Construction of the STC Model Inspired by ETAS

In this section, we illustrate how the STC model is constructed inspired by the ETAS model by comparing and analyzing the details of the STC model with the ETAS model one by one, as shown in Table I.

First, for the background earthquakes, in the ETAS model, $\mu(x, y)$ represents the background intensity in space continuous, while in the STC model, the 2-D matrix $\mu[x, y]$ represents the expected number of background earthquakes in space discrete.

Second, for the expectations of the number of aftershocks, in the ETAS model, as shown in (13), $k(m_i)$ indicates the expected number of earthquakes with magnitude m_i to produce

aftershocks, where M_0 is the minimum magnitude that can generate aftershocks and K and a denote the parameters to be fit. Correspondingly, in the STC model, as shown in (14), an input data matrix $I[i, j, k]$ represents the capability of aftershocks produced by earthquakes in that bin. In this process, the ETAS and STC model M_0 are equal to M_c and 0, respectively.

Third, for the probability distribution function of the time and location of the appearance of the aftershocks, denoted by a continuous time-space kernel function $g(t - t_i, x - x_i, y - y_i, m_i)$ in the ETAS model, as shown in (15). The parameters c and ω control the temporal decay, while d , γ , and ρ control the spatial decay. In the STC model, as shown in (16), turn time-space continuous functions $g(t - t_i, x - x_i, y - y_i, m_i)$ into the discrete 3-D convolution kernels $C[t - i, x - j, y - k]$. Each value in the convolution kernel is calculated from the gridded spatio-temporal distance. c and p and d and q are neurons with an initial value equal to 1, which are used to learn the pattern of aftershocks decaying over time and space after the earthquake occurs, respectively.

To summarize the above, in the ETAS model, the convolution operation of $k(m_i)$ and $g(t - t_i, x - x_i, y - y_i, m_i)$ in continuous time-space is employed to compute the triggered intensity for point (t, x, y) , whereas in the STC model, the convolution operation of $I[i, j, k]$ and $C[t - i, x - j, y - k]$ in discrete time-space is employed to compute the expected number of triggered earthquakes for bin $[t, x, y]$. The former is known as the Hawkes process or self-exciting process, while the latter is known as the convolutional operation.

However, in contrast to the traditional convolutional kernel, the advantage of the convolutional kernel in STC model is the ability to incorporate statistical prior knowledge, without many trainable variables. Instead, they are calculated via four neurons (d, q, c , and p) that are intricately linked with

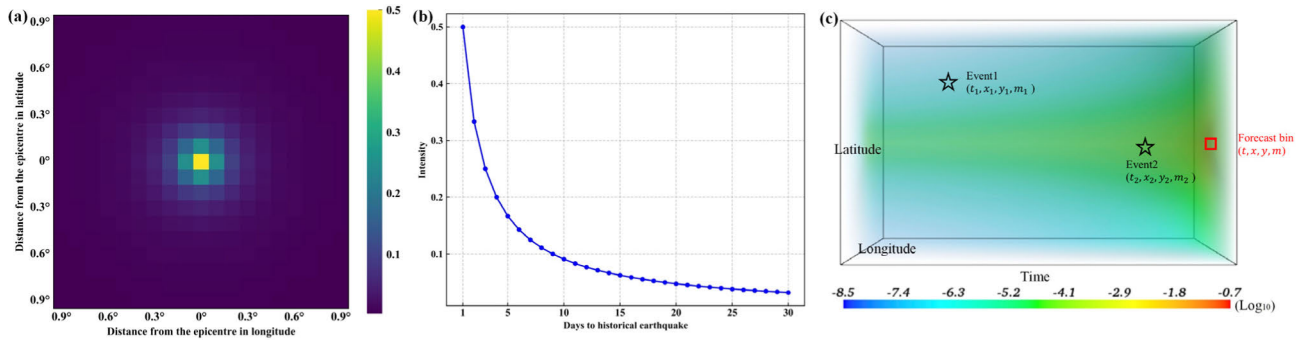


Fig. 3. (a) and (b) Spatial kernel function ($d = 1$ and $q = 1$) and temporal kernel function ($c = 1$ and $p = 1$), respectively. (c) Spatio-temporal kernel function ($d = 1$, $q = 1$, $c = 1$, and $p = 1$). The center to the right is the position of the spatio-temporal bin to be predicted. The color bar indicates the value of each grid in the convolution kernel.

TABLE II
OPTIMIZATION FLOW OF THE STC MODEL

Input: The processed dataset, including input data $I[t, x, y]$, true number $N[t, x, y, m]$ and true label $L[t, x, y, m]$.
1: Initialize the neurons $\alpha, \beta, c, p, d, q$ to 1.
2: Set counting notation $l = 0$ and initialize the background intensity $\mu[x, y]^{(0)}$ to 1.
3: Calculate $\Lambda[t, x, y, m]$ from Eq. (8) for each bin in the training and validation dataset.
4: Calculate $loss$ from Eq. (20) for each epoch in the validation dataset and record it as $loss^{(l)}$
5: Calculate $\hat{\mu}[x, y]$ for each epoch and record it as $\mu[x, y]^{(l+1)}$ in the training dataset, through the expected number of total earthquakes and triggered earthquakes.
6: If $\max \mu[x, y]^{(l+1)} - \mu[x, y]^{(l)} > \varepsilon$ and $loss^{(l)} < \min(loss^{(l-1)}, loss^{(l-2)}, loss^{(l-3)})$.
7: Replace $\mu[x, y]^{(l)}$ with $\mu[x, y]^{(l+1)}$ and optimize the STC model with gradient feedback only using the training dataset, then set $l = l + 1$ and go to step 3.
8: else
9: Stop training, take $\mu[x, y]^{(l+1)}$ as the background intensity and output the parameters $\alpha, \beta, c, p, d, q$.
10: End if.

spatio-temporal decay functions. This allows the construction of a convolutional kernel with statistical knowledge with a minimal number of parameters and significantly reduces the risk of overfitting that is often associated with conventional deep learning models.

Furthermore, there are two characteristics of the input data in the ETAS model: 1) which takes all historical earthquakes within the study area before the prediction time t and 2) the magnitude of the input earthquake often greater than M_c . However, the size of the input data for the STC model should be manually given in advance, which is called Ω . The size of the convolution kernel is also a hyperparameter, and it should be consistent with the size of the input data. For instance, when the input time of historical earthquakes is set to 30 days and the radius is set to 0.9° , a convolution kernel with the shape (19, 19, 30) will be constructed. Fig. 3(a) and (b) represents the exponential decay of aftershocks with the mainshock in space and time, respectively. Fig. 3(c) denotes the STC kernel obtained by multiplying both. The spatio-temporal bins to be predicted are indicated by the red squares, while the historical earthquakes are represented by the black stars. The triggered component of “Event 1” or “Event 2” for the predicted bin is determined by the spatial and temporal intervals ($|t_1 - t|$, $|x_1 - x|$, $|y_1 - y|$) or ($|t_2 - t|$, $|x_2 - x|$, $|y_2 - y|$) and the magnitude m_1 or m_2 , respectively. Historical earthquakes closer in time and space to the forecast bin have a greater triggered effect with the same magnitude.

Therefore, the optimization parameters of the ETAS model are $\mu, K, a, c, \omega, d, \gamma, \rho$, while the optimization neurons of the STC model are $\alpha, \beta, d, q, c, p$. To optimize these parameters or neurons, the maximum likelihood estimates were applied in the ETAS model, and the mean square error loss function was applied in the STC model.

During the optimization process, the 2-D matrix $\mu[x, y]$ of the STC model is updated, and the algorithm for updating it is derived from [38]. After each training epoch, a new estimate of the 2-D matrix $\mu[x, y]$ is obtained to replace the previous one. Until the 2-D matrix $\mu[x, y]$ and mean square error loss converge, the model stops training and calculates $\Lambda(t, x, y, m)$. The optimization algorithm flow for the STC model is shown in Table II. Moreover, to provide probabilities that at least one earthquake above target magnitude will occur in the next day, for the ETAS model, we conducted 10000 simulations to calculate $P_{ETAS}(t, x, y, m)$ based on the Gutenberg–Richter law, for the STC model, and $P_{STC}[t, x, y, m]$ will be calculated based on the assumption that earthquake occurrences follow a Poisson distribution.

D. Evaluation Metrics

To evaluate the performance of the ETAS and STC model, several indices have been defined as follows.

True Positive (TP): The count of actual positive samples is correctly identified by the model.

True Negative (TN): The count of actual negative samples is accurately classified as such by the model.

False Positive (FP): The count of actual negative samples is incorrectly classified as positive by the model.

False Negative (FN): The count of actual positive samples is erroneously classified as negative by the model.

Our evaluation metrics are given as follows.

1) **ROC Curves and Area Under the Curve:** The ROC curve is a standard tool for evaluating binary classifiers, plotting the FP rate (FPR) against the TP rate (TPR) across various thresholds. The formulas for FPR and TPR are $FPR = FP/(FP + TN)$ and $TPR = TP/(TP + FN)$, respectively. The area under the curve (AUC) quantifies the model's overall performance, with a higher AUC indicating superior performance. An AUC of 0.5 corresponds to a Poisson baseline model, while an AUC approaching 1 signifies excellent performance.

2) **PR Curves and AUC(PR):** PR curves provide a more informative evaluation result in the class imbalance task [39], [40]. These curves plot precision $[TP/(TP + FP)]$ against recall $[TP/(TP + FN)]$. The AUC(PR) represents the area under the PR curve, with a random baseline expectation of a horizontal line intersecting the vertical axis at the positive sample rate. A model outperforms the random baseline if the AUC(PR) exceeds this rate, with a larger difference indicating better performance. The PR curve differs from the ROC curve in its evaluation metrics and sensitivity to class imbalance.

3) **Parimutuel Gambling Score:** While ROC and PR curves are effective for ranking models, they compress model outputs linearly from $[0, 1]$ to $[0, 0.01]$, which may not reflect the significance of small probability differences in earthquake prediction. To address this, we apply the PGS, as proposed by Zechar and Zhuang [41] and tested by Serafini et al. [42]. The PGS compares the probabilistic predictions of two models, p_1 and p_2 , by calculating scores based on the difference in predicted probabilities. For earthquakes that occur, the scores for the first model and the second model are $(p_1 - p_2)/(p_1 + p_2)$ and $(p_2 - p_1)/(p_2 + p_1)$, respectively. For nonoccurrences, p_1 and p_2 are simply replaced by $1 - p_1$ and $1 - p_2$, respectively. The score is proportional to the difference between the probabilities given by the models, and the larger the difference, the higher the score for the winning model.

Other key performance metrics, including confusion matrix and F1-score $[2TP/(2TP + FP + FN)]$, were calculated to provide a comprehensive evaluation of the STC model.

E. Implementation Details

The code of the STC model is implemented in Python 3.6 using TensorFlow [43]. The computations are conducted on a desktop PC equipped with an NVIDIA GeForce RTX 2080 GPU and a Xeon¹ W-2102 CPU with 128-GB memory.

To evaluate the performance of the STC model during training and to monitor potential overfitting or underfitting, the loss decline curves were plotted for both the training and validation sets, as shown in Fig. 4. The STC model demonstrated convergence at the tenth epoch, accompanied by a lack of fluctuations in training and validation loss. This indicates that

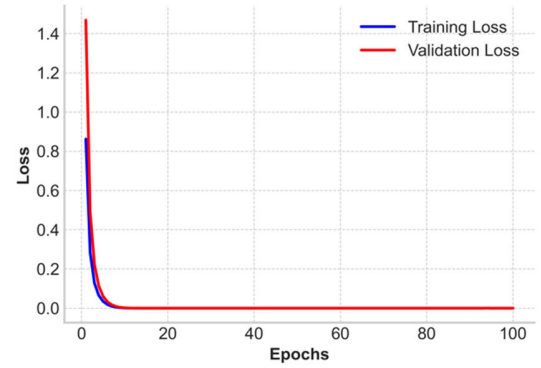


Fig. 4. Training and validation loss (the input magnitude and target magnitude both are 3.0, and the input time and radius are 90 days and 0.9° , respectively).

the model was able to fit the training data effectively, without exhibiting signs of overfitting or underfitting. This further substantiates the rationale for employing a model comprising only six neurons. Therefore, the early stopping mechanism is employed in the training process to improve the efficiency. Furthermore, to provide a comprehensive understanding of the computational efficiency under the current configuration, we recorded the training and inference times. The average training time per epoch is about 60 s, and the total training time amounts to about 10 min. For inference, the model requires 5 s per prediction.

IV. RESULTS

A. Impact of Spatio-Temporal Volume of Input Data on the Classification Performance of STC Models

For the STC model, the spatio-temporal volume of the historical earthquake is set by varying the input times {10, 20, 30, 40, 50, 60, 70, 80, and 90 days} and input radius $\{0.3^\circ, 0.6^\circ, 0.9^\circ, 1.2^\circ, 1.5^\circ, 1.8^\circ, \text{ and } 2.1^\circ\}$. The ETAS model, in contrast, employed the entire historical earthquake catalog for parameter optimization. First, the magnitude threshold for input historical earthquakes of the STC model is set the same as the ETAS model ($M \geq 3$). It allows for a fairer comparison of the performance of the STC model with the ETAS model, besides an exploration of the impact of the spatio-temporal volume of the input data on the performance of the STC model. Each performance was evaluated with the AUC(ROC) and AUC(PR) metrics for predictive magnitudes $M \geq 3$, $M \geq 4$, and $M \geq 5$.

The AUC(ROC) for ETAS with the predictive magnitudes $M \geq 3$, $M \geq 4$, and $M \geq 5$ was 0.919, 0.879, and 0.744, respectively, and the corresponding AUC(PR) values were 0.02543, 0.00759, and 0.00087. The AUC(ROC) and AUC(PR) for the STC model with different input spatio-temporal volumes are illustrated in Fig. 5. The AUC(ROC) increased significantly with the initial expansion of input time and radius. However, the increasing rate diminished after reaching 60 days and 0.9° . The AUC(PR), which provides a more nuanced evaluation in imbalanced datasets, showed only slight increases for $M \geq 3$ with the STC model, with minimal variation for $M \geq 4$ or 5. This suggests that extending the input spatio-temporal volume can improve the overall classification performance. However, the improvement

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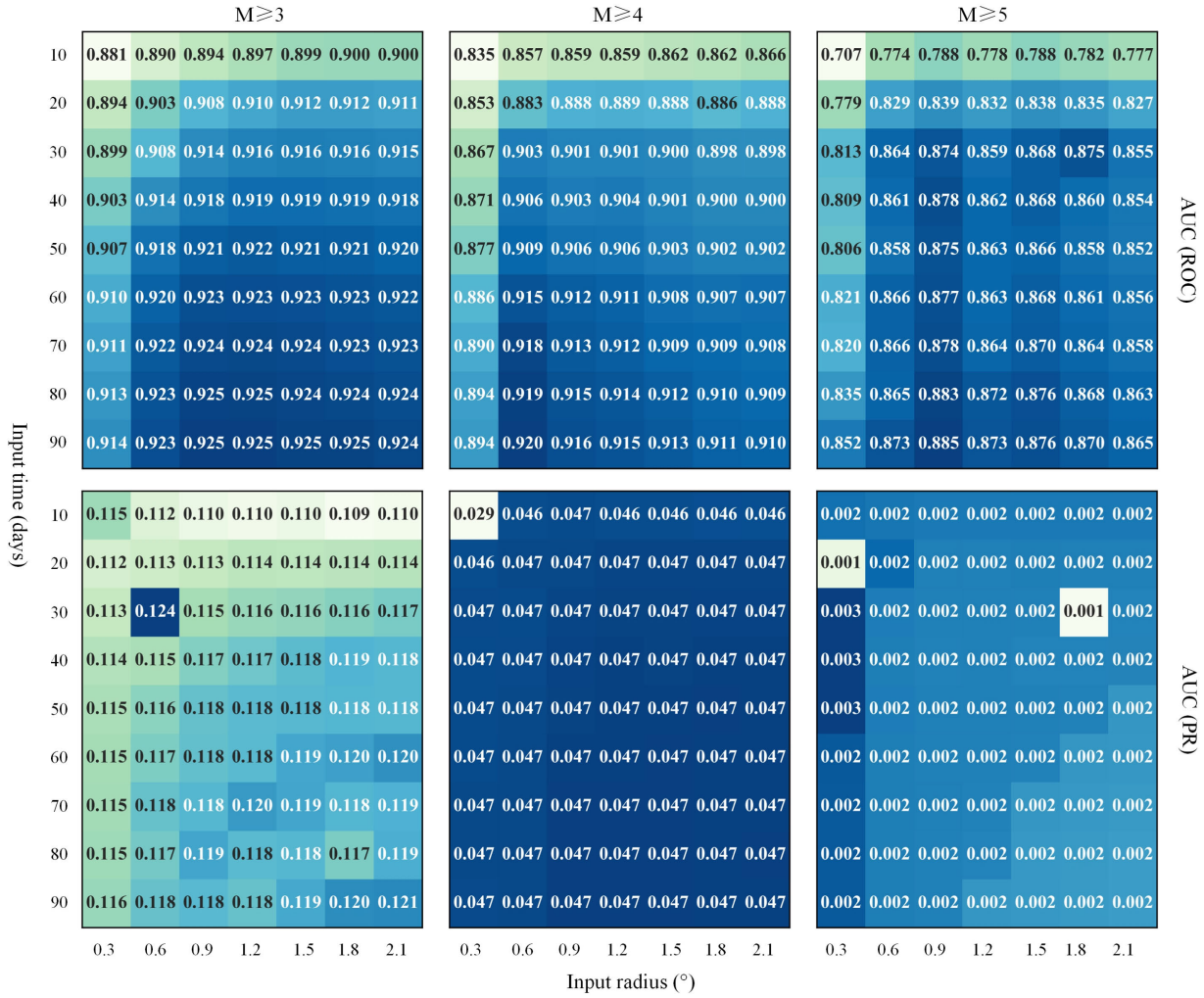


Fig. 5. AUC(ROC) and AUC(PR) of the STC model based on the test dataset for different input times and input radius.

in positive sample classification is not significant. Moreover, processing a large spatio-temporal input volume requires more computational cost and time. Thus, selecting a balanced spatio-temporal range for input data ensures both effective prediction and operational efficiency. In the competition, the STC model outperformed ETAS across all target magnitudes both in AUC(ROC) and AUC(PR) metrics.

The STC model in this study considers up to 90 days and 2.1° of historical data due to the limitation of computational capability, yet the effects of significant earthquakes may persist beyond this volume. Future research could explore the impact of longer time series of large earthquakes by increasing computational resources. In the following analysis, the STC model with an input spatio-temporal volume of 90 days and 0.9° will be employed to explore what the STC model learned and to compare its advantages over ETAS using ROC curves, PR curves, and PGS.

B. Parametric Analysis of the STC Model

In contrast to traditional deep learning models, which often lack interpretability, the STC model offers insights into the patterns it has learned from the perspective of the triggered and background earthquakes. The STC model is optimized by six

trainable neurons: $\alpha = 1.504$, $\beta = 0.005$, $c = -0.160$, $p = 1.458$, $d = 0.202$, and $q = 1.474$. α and β represent the relative weights assigned to triggered earthquakes and background earthquakes, respectively. c and p and d and q are responsible for determining the decay rate of the spatio-temporal kernel in time and space, respectively. A decrease in c and d enhances the model's sensitivity to triggered earthquakes in the surrounding spatial-temporal units, while an increase in p and q indicates a faster attenuation rate.

In Fig. 6, we present a visualization of the true label and output probabilities for both ETAS and STC models. In Fig. 6(a), earthquakes have significant spatial clustering in California, and the spatial distributions of the output probabilities of both ETAS and STC models are similar to it, as shown in Fig. 6(b) and (c), respectively. The output probability of STC is composed of radial triggered component [Fig. 6(d)] with scattered background component [Fig. 6(e)]. The spatial distribution of positive samples in the training dataset is shown in Fig. 6(f). The background earthquakes learned by the STC model reflect the spatial distribution of positive samples in the training dataset. This suggests that the model identifies regions with a higher frequency of earthquakes in the training data as more prone to future earthquake activity.

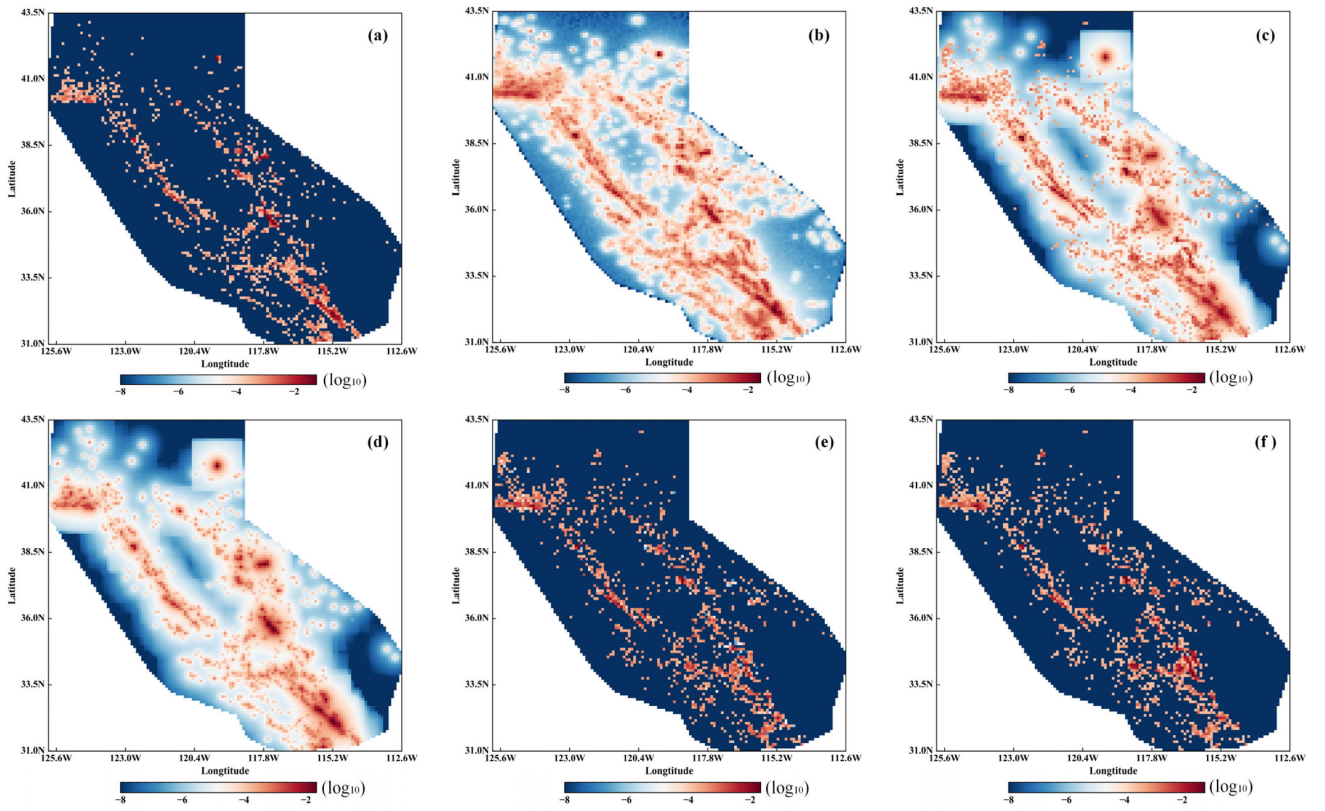


Fig. 6. (a) Spatial distribution of the positive sample rate in the test dataset. (b) and (c) Spatial distribution of the output probabilities of the ETAS model and the STC model in the test dataset, respectively. (d) and (e) Spatial distribution of the triggered and background component provided by the STC model, respectively. (f) Distribution of the positive sample rate in the training dataset. In the above plots, any values less than 10^{-8} have been replaced with 10^{-8} .

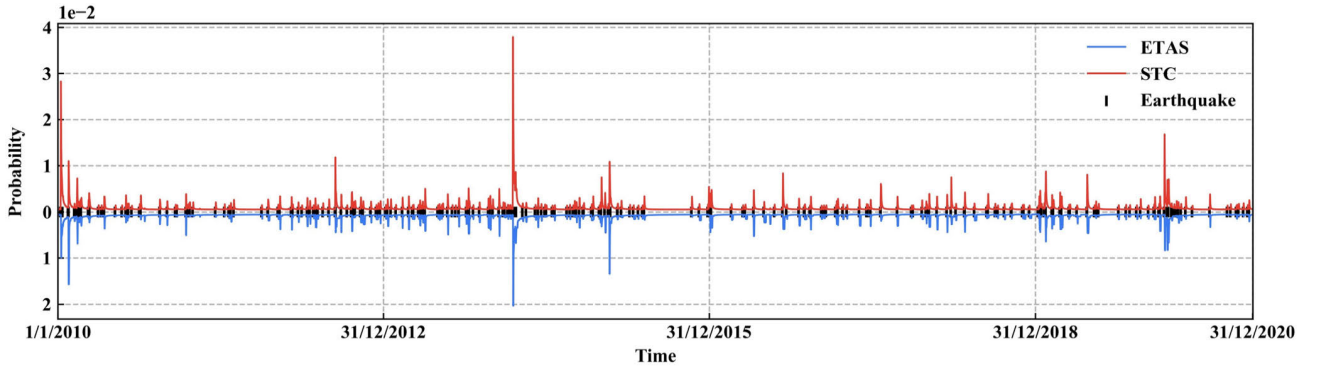


Fig. 7. ETAS (blue line) and STC (red line) provide the average probability over time for a specific region (125.9°W–123.9°W, 40°N–41°N) based on test datasets ($M \geq 3$). The short black line indicates the occurrence of at least one earthquake ($M \geq 3$) in the region at that moment.

In addition, this study presents a temporal analysis of the STC and ETAS models' responses to earthquake activity in a confined region, as shown in Fig. 7. During periods of no earthquake, both models yielded low background probabilities for this region. However, they exhibited a significant increase in aftershock prediction probabilities after mainshocks occurred. The ability of the models to adjust their predictions dynamically in response to mainshocks is highlighted by this temporal response.

C. Comparison of STC and ETAS Models Based on Evaluation Metrics

The comparative analysis of the STC and ETAS models' performance is presented in Fig. 8, based on the ROC and PR curves. As the proportion of positive samples is much larger

than that of negative samples, logarithmic scaling is applied to provide a clearer depiction of classification performance. The STC model demonstrates significantly better early stage classification performance for $M \geq 3$. This is evidenced by a lower FPR at equivalent TPR, as shown in Fig. 8(a). ETAS surpasses STC in terms of FPR only when the TPR approaches 0.9, indicating that STC outperforms ETAS for predicting approximately the first 90% of earthquakes, while ETAS is marginally superior for predicting the final 10%. Fig. 8(b) and (c) depicts the ROC curves for $M \geq 4$ and $M \geq 5$, respectively, where STC maintains a clear advantage in the initial classification stages.

The PR curve, which is more informative in imbalanced datasets, is also displayed with logarithmic scaling for the recall axis to display early stage performance. In Fig. 8(d), for

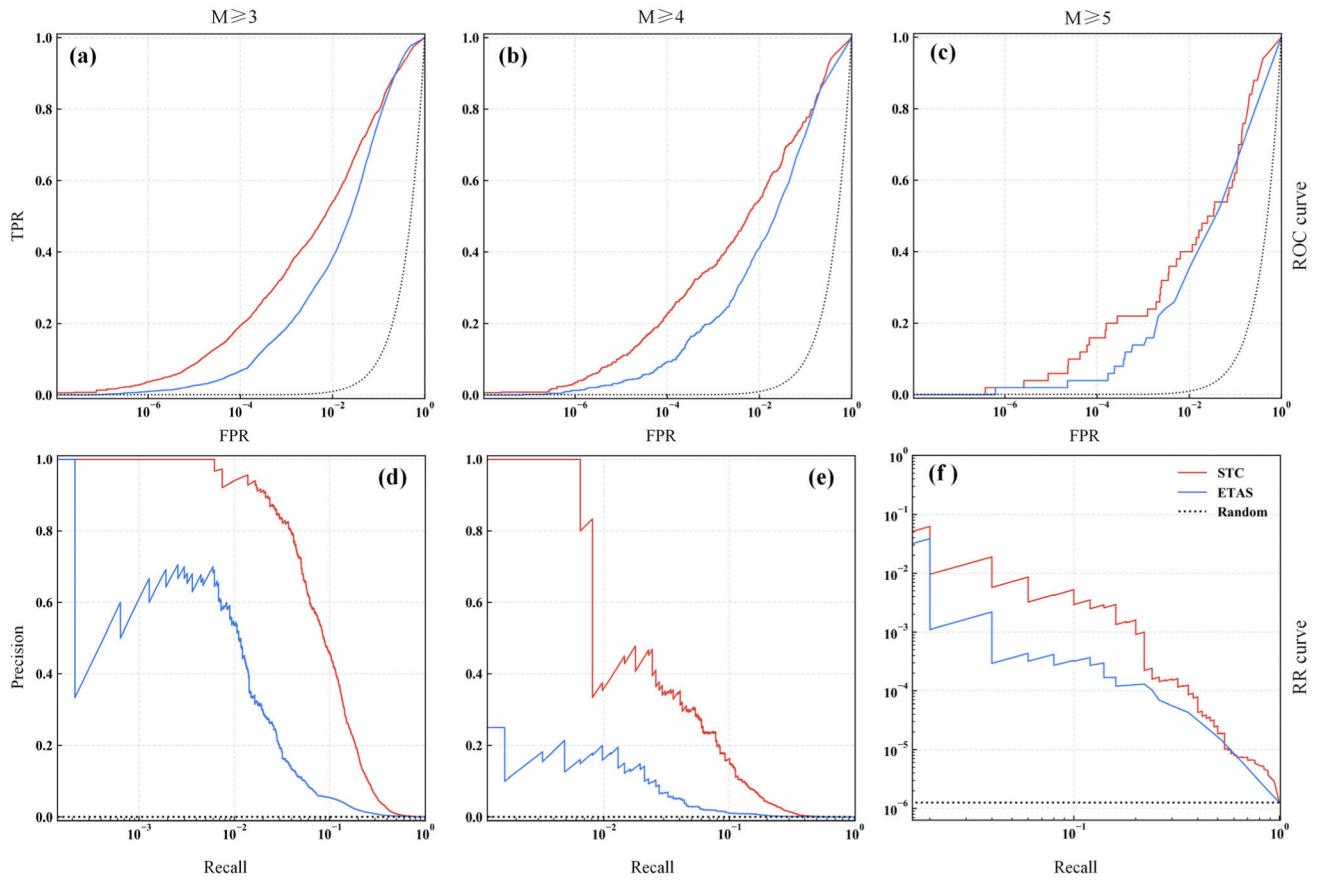


Fig. 8. (a)–(c) ROC curve of ETAS (blue line) and STC (red line) based on the test dataset for $M \geq 3$, $M \geq 4$, and $M \geq 5$, respectively. (d)–(f) PR curve of ETAS (blue line) and STC (red line) based on the test dataset for $M \geq 3$, $M \geq 4$, and $M \geq 5$, respectively. The AUC(ROC) of STC for the three target magnitudes is 0.925, 0.916, and 0.885. The AUC(ROC) of ETAS for the three target magnitudes is 0.919, 0.879, and 0.744. The AUC(PR) of STC for the three target magnitudes is 0.118, 0.047, and 0.002. The AUC(PR) of ETAS for the three target magnitudes is 0.025, 0.008, and 0.0009. The random baseline for AUC(ROC) is 0.5, and for AUC(PR), it is 0.0001, 0.00002, and 0.000001 for the three target magnitudes.

	$M \geq 3$		$M \geq 4$		$M \geq 5$	
Actual Negative	34420514	5457465	33348203	6533849	31697612	8185006
Actual Positive	732	3957	109	507	8	42
	Predicted Negative	Predicted Positive	Predicted Negative	Predicted Positive	Predicted Negative	Predicted Positive

Fig. 9. Confusion matrix for the STC model based on test set. The input magnitude and target magnitude both are 3.0, and the input time and radius are 90 days and 0.9° , respectively.

$M \geq 3$, STC achieves a precision exceeding 90% for the top 1% of earthquakes and over 40% precision at a recall rate of 10%, significantly outperforming ETAS. The precision of both models declines rapidly as recall increases due to the dominance of negative samples. Fig. 8(e) and (f) presents the PR curves for $M \geq 4$ and $M \geq 5$, respectively. Predicting positive samples becomes increasingly challenging with higher target magnitudes. Nonetheless, STC maintains higher precision in the early prediction stages, and its AUC(PR) is substantially superior to ETAS across all three target magnitudes.

To further evaluate the classification performance of the STC model, we analyzed confusion matrices for three different magnitude thresholds: 3.0, 4.0, and 5.0, as shown in Fig. 9.

The metrics derived from these confusion matrices provide insight into the model's ability to correctly identify both positive (earthquakes of the target magnitude) and negative (nonearthquake or earthquakes below the target magnitude) instances.

For $M \geq 3$, the model achieved a TP count of 3957, a TN count of 34 420 514, an FP count of 5 457 465, and an FN count of 732. This corresponds to a TPR of 0.8439, an FPR of 0.1369, a precision of 0.0007, and an $F1$ -score of 0.0014. These metrics indicate a relatively high sensitivity, but a low precision and $F1$ -score, suggesting that while the model detects many earthquakes, it also produces a substantial number of FPs. However, this is due to too fine a spatio-temporal resolution ($0.1^\circ \times 0.1^\circ \times 1$ day).

For $M \geq 4$, a TP count of 507, a TN count of 33 348 203, an FP count of 6 533 849, and an FN count of 109. At this threshold, the TPR is 0.8231, the FPR is 0.1638, the precision is 0.00008, and the $F1$ -score is 0.0002. This shift reveals a decrease in TPs and an increase in FPs, highlighting a tradeoff where the model becomes more conservative, potentially missing more actual earthquakes while still generating a significant number of false alarms.

For $M \geq 5$, the model's performance shows a further decline in sensitivity, with a TP count of 42, a TN count of 31 697 612, an FP count of 8 185 006, and an FN count of 8. The corresponding TPR is 0.84, the FPR is 0.2052, the precision

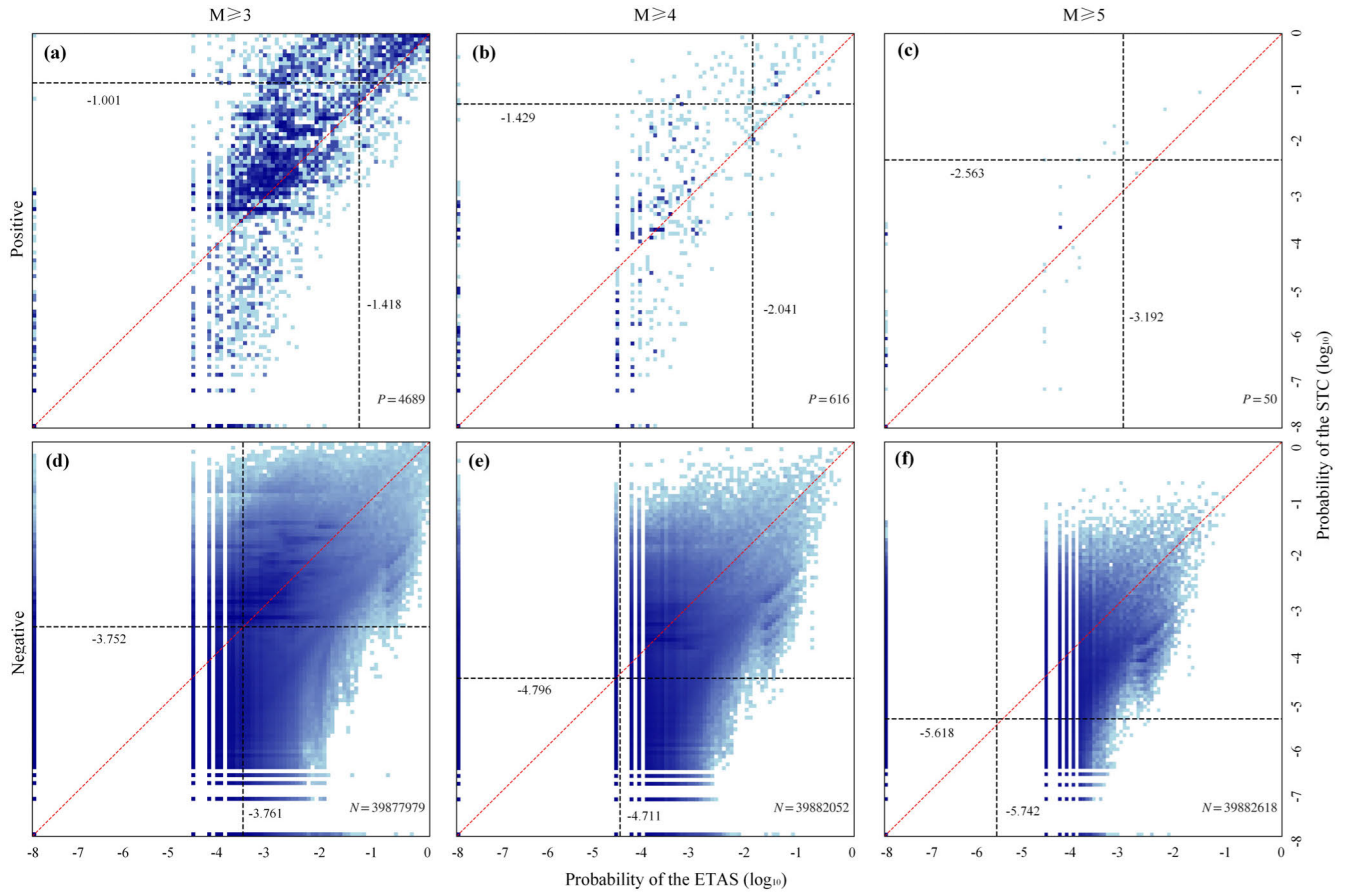


Fig. 10. (a)–(c) Probability distribution output of ETAS versus STC for positive samples. (d)–(f) Probability output of ETAS versus STC for negative samples. The red diagonal line indicates 1:1, and the black dashed lines parallel to the horizontal and vertical axes indicate the mean probability of the STC and ETAS, respectively. “P” and “N” denote the number of positive and negative samples in the test dataset, respectively. The input magnitude, time, and radius were set to 3.0, 90 days, and 0.9° , respectively. In the above plots, any values less than 10^{-8} have been replaced with 10^{-8} .

is 0.000005, and the $F1$ -score is 0.00001. This configuration indicates that the model is extremely conservative, with a very high FPR and a lower TPR. The significant increase in FPs and the reduction in TPs suggest that the model’s efficacy in detecting higher magnitude earthquakes is limited, potentially due to the sparsity of such earthquakes.

Beyond the classification performance evaluation, we have further assessed the probability forecasting capabilities of the STC and ETAS models through the PGS. Fig. 10(a)–(c) demonstrates that the STC model assigns a higher average probability to positive samples compared to ETAS under the three target magnitudes. Conversely, for negative samples, both models yield similar average probabilities, as indicated in Fig. 10(d)–(f). Despite the scarcity of positive samples, the STC model achieves a higher cumulative PGS for the three target magnitudes: 1154.848, 143.096, and 24.085. Given the substantial disparity between negative and positive samples, the STC model’s cumulative PGS for negative samples is -44.558 , 65.378 , and -12.062 . In summary, for $M \geq 3$ and $M \geq 5$, the STC model demonstrates a pronounced advantage in predicting positive samples, with a marginally weaker performance for negative samples compared to ETAS. For $M \geq 4$, STC has an advantage both in predicting positive samples and negative samples. The overall PGS for the STC model across the three target magnitudes is 1110.290, 208.474, and 12.023. These results suggest that the STC model’s probability

forecasting performance significantly outperforms that of the ETAS model.

V. DISCUSSION

A. Impact of Small-Magnitude Earthquakes on the STC Model

Small-magnitude earthquakes have the potential to provide valuable predictive information. Previous studies by Zlydenko et al. [32] and Stockman et al. [44] have used AUC(ROC) and maximum likelihood to demonstrate that NNs can enhance earthquake prediction performance with small-magnitude earthquakes. However, the ETAS model is less flexible in incorporating such data due to its rigid threshold [26]. In contrast, the STC model is not limited by this threshold during training, allowing us to investigate the benefits of incorporating smaller magnitude earthquakes based on AUC(ROC) and AUC(PR) metrics. In Fig. 11, we divided the input magnitude thresholds into 16 intervals from 0 to 3 with increments of 0.2, including only earthquakes above each threshold as STC model inputs.

In Fig. 11(a), for predicting earthquakes with magnitude $M \geq 5$, a clear trend in AUC(ROC), an initial increase followed by a decrease, that is, lowering the input magnitude threshold from 3.0 to 2.4, can significantly improve the overall classification performance of the STC model when the

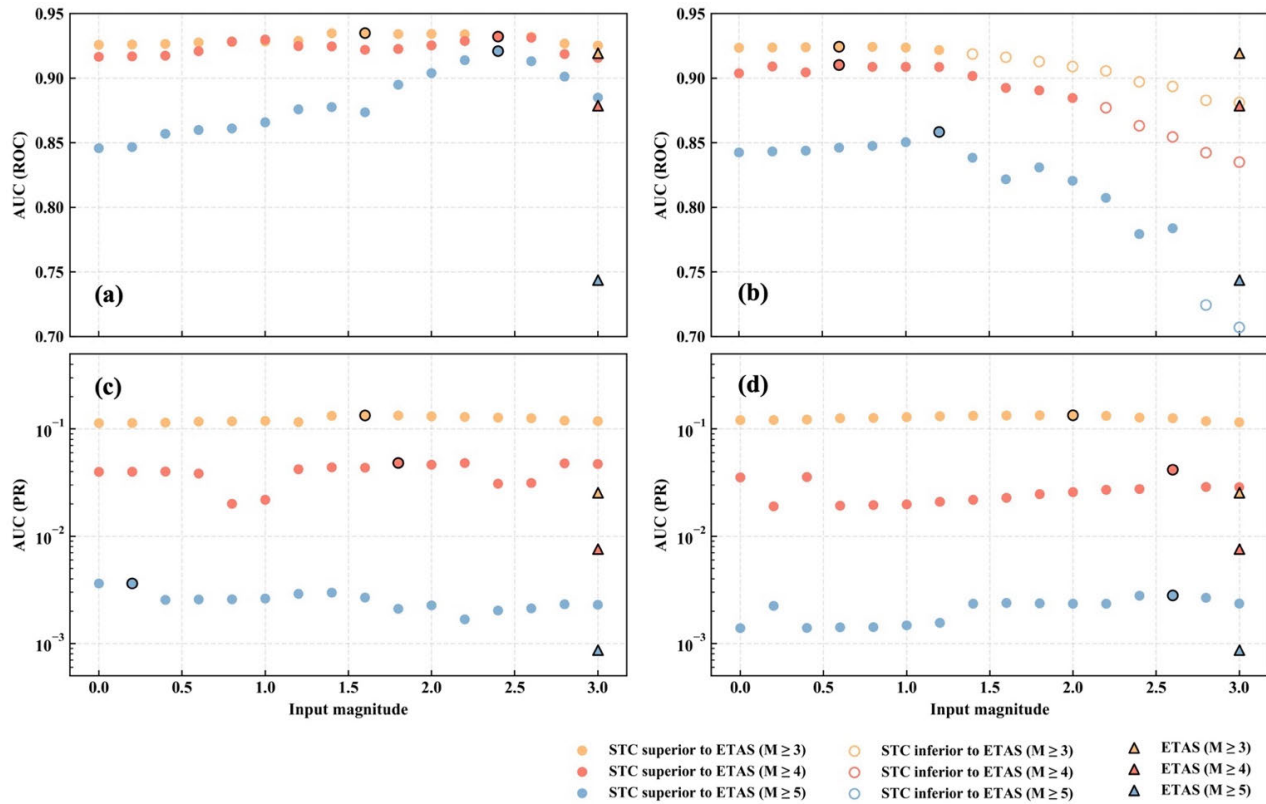


Fig. 11. Impact of input magnitude thresholds on STC model for different magnitudes earthquake prediction under different spatio-temporal ranges. The horizontal axis represents the magnitude threshold of the input earthquake, while the yellow, pink, and blue circles indicate that the forecasting magnitude is $M \geq 3$, $M \geq 4$, and $M \geq 5$, respectively. (a) and (c) AUC(ROC) varies with input magnitude threshold for input time and radius of 60 days and 0.9° and ten days and 0.3° , respectively. (b) and (d) AUC(PR) varies with input magnitude threshold for input time and radius of 60 days and 0.9° and ten days and 0.3° , respectively.

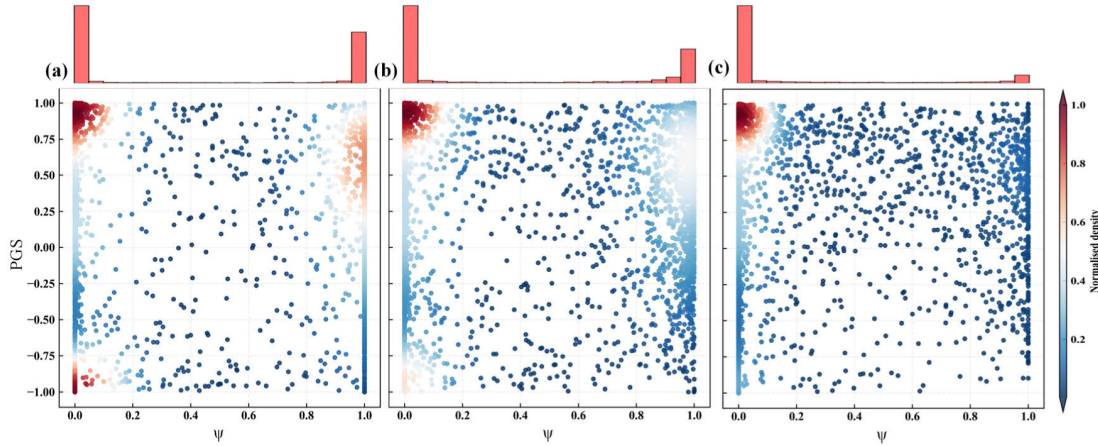


Fig. 12. Density scatter plot represents the relationship between ψ and the PGS for different input magnitude thresholds and input size. The histogram represents the distribution of ψ . (a) Input size is ten days, 0.3° , and the input magnitude threshold is 3. (b) Input size is 90 days, 0.9° , and the input magnitude threshold is 3. (c) Input size is ten days, 0.3° , and input magnitude threshold is 3.

spatio-temporal volume of the input data is 60 days and 0.9° . However, further lowering the magnitude threshold leads to a decrease in performance. Furthermore, there is no significant impact on the overall classification performance of the STC model for predicting earthquakes with magnitude $M \geq 3$ and $M \geq 4$.

However, when the spatio-temporal volume of the input data is reduced to ten days and 0.3° and the magnitude threshold is 3.0, as shown in Fig. 11(b), the AUC(ROC) of the STC model is inferior to the ETAS model for all three target magnitudes. However, lowering the magnitude threshold leads to a great

improvement for AUC(ROC). This makes the STC model surpassing ETAS and stabilizing at a higher level. Therefore, a lower magnitude threshold can compensate for the reduced spatio-temporal input range.

The AUC(PR) for STC does not significantly vary with the input magnitude, yet it remains significantly superior to ETAS, as illustrated in Fig. 11(c) and (d). In summary, the inclusion of small-magnitude earthquakes can improve the overall classification performance for the STC model, but it has a limited effect on the ability to classify positive samples.

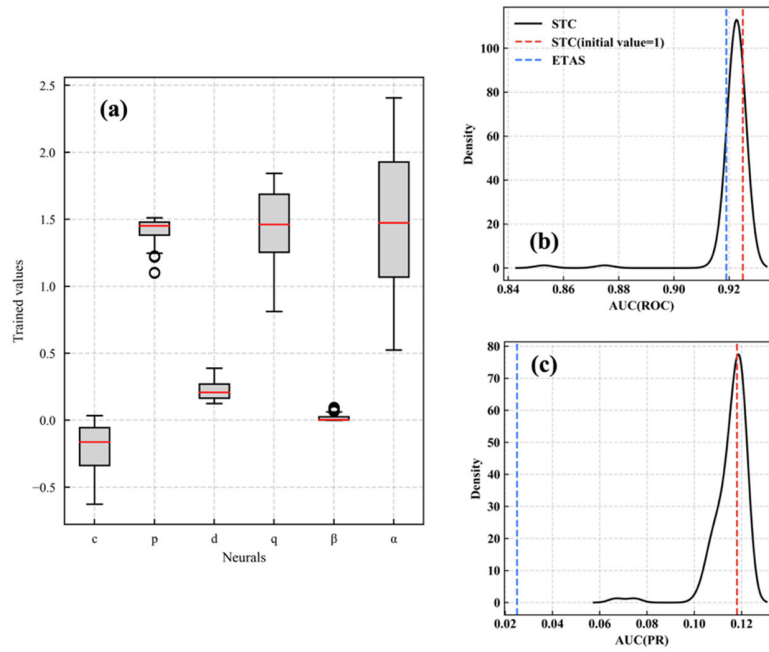


Fig. 13. (a) Distribution of final parameters at the end of training for 100 sets of the STC model with different initial values. (b) and (c) Distribution of AUC(ROC) and AUC(PR) for the STC model and ETAS model, respectively. The black solid line represents the distribution of the scores of the 100 sets of STC model with different initial values, and the red and blue dashed lines represent the scores of the STC (initial value = 1) and ETAS model, respectively.

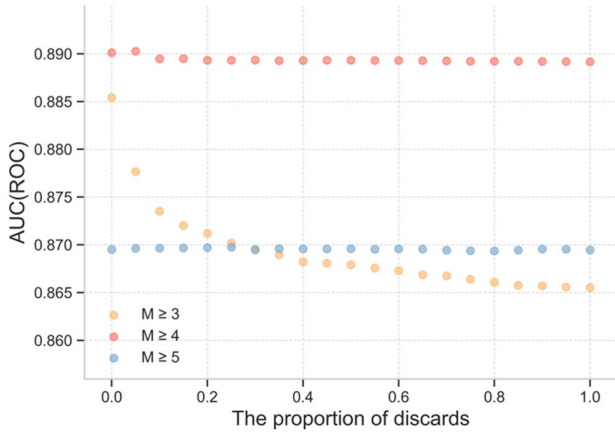


Fig. 14. Impact of the proportion of discards for small earthquakes ($M \leq 1$) in the training set on the performance of the STC model. The horizontal axis represents the proportion of discards, and the vertical axis represents the AUC(ROC) of the STC model on the test set. The yellow, pink, and blue circles indicate that the forecasting magnitude is $M \geq 3$, $M \geq 4$, and $M \geq 5$, respectively.

B. Advantages of the STC Model Over the ETAS Model in Triggered and Background Earthquakes

A comparative analysis was conducted to analyze the performance of the STC and ETAS models in predicting background and triggered earthquakes. In Fig. 12, the three subplots illustrate the different spatio-temporal volumes and magnitude of the input data. The horizontal axis denotes the independent probability (ψ) of each positive sample with magnitude $M \geq 3$ in the test dataset, which is calculated by the ratio of background component to total intensity. The vertical axis represents the PGS of the STC model in the betting against the ETAS model in the prediction of each positive sample.

The STC model is superior to the ETAS model significantly for low independence probability samples, i.e., located at the

left edge of the subplot. Furthermore, the STC model wins more PGS even with a high magnitude threshold and limited spatio-temporal input volume, as shown in the top-right corner of Fig. 12(a). This flexibility in utilizing background intensity is another key advantage of STC.

When the spatio-temporal volume from ten days and 0.3° increased to 60 days and 0.9° , as shown in the transition from Fig. 12(a) and (b), the percentage of triggered samples also increased. The advantage of the STC model is predicting the triggered earthquakes without over-reliance on the background intensity. Moreover, when reducing the input magnitude threshold from 3 to 0, roughly 90% of the earthquake samples have an independent probability of less than 10%, as shown in the transition from Fig. 12(a)–(c). Lowering the threshold of the input earthquake magnitude significantly increases the percentage of triggered earthquakes, despite the limited spatial and temporal range of the input. The distribution of PGS demonstrates that the STC model dominates the ETAS model in predicting triggered earthquakes. In conclusion, STC's superiority over ETAS is primarily evident in its prediction of triggered earthquakes, an advantage that can be further enhanced by the inclusion of appropriately selected small-magnitude earthquakes as model inputs.

C. Effect of the Initial Parameter

The different initial values set for both the ETAS model and deep learning models will impact the parameters fitting. Therefore, this section aims to investigate the impact of initial values on the training of the STC model and to determine whether the superiority of the STC model over the ETAS model is statistically significant. Using the example of input data with a spatio-temporal volume of 60 days, 0.9° , and predicted magnitude threshold set to 3, 100 sets of initial values were randomly set between 0.5 and 1.5 with uniform distribution for the six neurons in the STC model. Fig. 13(a)

represents the distribution of the final values of the six neurons for the 100 sets of STC models upon completion of training. Compared to the initial values, the neurons p , q , and α exhibit an overall increasing trend, while the neurons c , d , and v show an overall decreasing trend. However, neurons p , d , and v are less affected by the initial values and converge to a narrower range, while neurons c , q , and α are relatively less stable. Regarding performance evaluation, as shown in Fig. 13(b) and (c), the scores of the 100 sets of STC models exhibit a normal distribution overall. Ninety-seven sets of STC models have better AUC(ROC) than the ETAS model, and all STC models have better AUC(PR) than the ETAS model. This indicates that the initial parameter values have less influence on the training of STC models and the classification performance of STC models is superior to that of ETAS models not by chance.

D. Effect of the Small Earthquake in the Training Set

There is a significant absence of small earthquakes ($M \leq 0.5$) between 1/1/1990 and 12/31/1999, as shown in Fig. 1(c). Therefore, the effect of small earthquakes in the training set on the STC model is explored. Specifically, the earthquake catalog for the period between 1/1/2000 and 12/31/2006 was employed as the training set, the period between 1/1/2007 and 12/31/2013 was designated as the validation set, and the period between 1/1/2014 and 12/31/2020 was designated as the test set. Then, a proportion of earthquakes with $M \leq 1$ in the training set is randomly discarded to explore the effect of the proportion of discards on the performance of the STC model. To illustrate the methodology, we will consider the example of input data with a spatio-temporal volume of 60 days, 0.9° . An intriguing outcome was observed whereby the exclusion of small earthquakes from the training set had a negligible impact on the predictive efficacy of the STC model when the target magnitude with $M \geq 4$ or $M \geq 5$. Conversely, when the target magnitude with $M \geq 3$, the performance of the STC model appeared to decline significantly with the increase in the proportion of minor earthquakes discarded, as illustrated in Fig. 14. Therefore, it is recommended that the effects of small earthquakes are not be ignored in the input data if the target magnitude is also small.

VI. CONCLUSION

By integrating prior knowledge of ETAS model that aftershocks decay in time and space after the main shock into CNNs, the STC model was proposed to improve the performance and efficiency of earthquake forecasting. The STC model outperformed the ETAS model in next-day earthquake forecasting with magnitudes $M \geq 3$, $M \geq 4$, and $M \geq 5$. Furthermore, the inclusion of earthquakes below the completeness magnitude threshold has been shown to significantly boost the overall predictive performance. In contrast to complete deep learning approaches, the STC model benefits from the incorporation of prior knowledge, which not only enhances interpretability but also mitigates the risks of underfitting and overfitting.

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Chengxiang Zhan received the B.E. degree from Qinghai University, Qinghai, China, in 2019, and the M.S. degree from China University of Geosciences (Beijing), Beijing, China, in 2022, where he is currently pursuing the Ph.D. degree.

His main research interests include deep learning, statistical seismology, and remote sensing.



Shichen Gao received the B.S. degree in mathematics from Shandong University, Shandong, China, in 1985, and the Ph.D. degree from the Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing, China, in 2008.

His main research interests include applied mathematics and computer science.



Ying Zhang received the B.S. degree from Wuhan University, Wuhan, China, in 2016, and the Ph.D. degree from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2021.

From 2021 to 2023, he was a Post-Doctoral Fellow with the Department of Geophysics, School Earth and Space Sciences, Peking University, Beijing. He is currently an Associate Professor with the University of Science and Technology Beijing, Beijing.

His research interests include thermal infrared remote sensing and machine learning.



Jiawei Li received the B.S. degree from Chengdu University of Technology, Chengdu, China, in 2014, the M.S. degree from the Institute of Geophysics, China Earthquake Administration, Beijing, China, in 2017, and the Ph.D. degree from Peking University, Beijing, in 2020.

From 2020 to 2023, he was a Post-Doctoral Researcher at the Institute for Risk Analysis, Prediction, and Management (RisksX), Southern University of Science and Technology, Shenzhen, China, where he is currently a Senior Researcher.

His main research interests include statistical seismology and real-time seismology.



Qingyan Meng received the B.S. and M.S. degrees from Northeast Agricultural University, Harbin, China, in 1993 and 1996, respectively, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 1999.

He is currently a Professor at the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His main research interests include remote sensing and deep learning.